$$\dfrac{.}{B\mathcal{R}G, D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G}\ \text{identity}$$

$$\dfrac{.}{D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash D\mathcal{R}B, B \Vdash A, G \Vdash A, B\mathcal{R}G}\ \text{identity}$$

$$\dfrac{.}{D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash D\mathcal{R}G, B \Vdash A, G \Vdash A, B\mathcal{R}G}\ \text{identity}$$

andR

$$D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash (D\mathcal{R}B) \wedge (D\mathcal{R}G), B \Vdash A, G \Vdash A, B\mathcal{R}G$$

implyL

$$((D\mathcal{R}B) \wedge (D\mathcal{R}G)) \to (B\mathcal{R}G), D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

forallL

$$\forall v.(((D\mathcal{R}B) \wedge (D\mathcal{R}v)) \to (B\mathcal{R}v)), D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

forallL

$$\forall u.(\forall v.(((D\mathcal{R}u) \wedge (D\mathcal{R}v)) \to (u\mathcal{R}v))), D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

forallL

$$\forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))), D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

shuffleL

$$B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))), D\mathcal{R}B, D\mathcal{R}G \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

shuffleL

$$D\mathcal{R}G, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))), D\mathcal{R}B \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

shuffleL

$$D\mathcal{R}B, D\mathcal{R}G, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash B \Vdash A, G \Vdash A, B\mathcal{R}G$$

implyR

$$D\mathcal{R}G, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash (D\mathcal{R}B) \to (B \Vdash A), G \Vdash A, B\mathcal{R}G$$

existsR

$$D\mathcal{R}G, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A)), G \Vdash A, B\mathcal{R}G$$

exchangeR

$$D\mathcal{R}G, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash G \Vdash A, \exists E.((D\mathcal{R}E) \to (E \Vdash A)), B\mathcal{R}G$$

implyR

$$B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash (D\mathcal{R}G) \to (G \Vdash A), \exists E.((D\mathcal{R}E) \to (E \Vdash A)), B\mathcal{R}G$$

existsR

$$B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A)), \exists E.((D\mathcal{R}E) \to (E \Vdash A)), B\mathcal{R}G$$

contractionR

$$B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A)), B\mathcal{R}G$$

exchangeR

$$B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash B\mathcal{R}G, \exists E.((D\mathcal{R}E) \to (E \Vdash A))$$

$$\dfrac{.}{D\mathcal{R}G, G \Vdash A, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) -}$$

$$G \Vdash A, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v))$$

$$G \Vdash A, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash$$

$$(B\mathcal{R}G) \to (G \Vdash A), B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A))$$

existsL

$$\exists F.((B\mathcal{R}F) \to (F \Vdash A)), B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A))$$

mDiamondL

$$B \Vdash \Diamond A, B\mathcal{R}D, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A))$$

exchangeL

$$B\mathcal{R}D, B \Vdash \Diamond A, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \exists E.((D\mathcal{R}E) \to (E \Vdash A))$$

mDiamondR

$$B\mathcal{R}D, B \Vdash \Diamond A, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash D \Vdash \Diamond A$$

implyR

$$B \Vdash \Diamond A, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash (B\mathcal{R}D) \to (D \Vdash \Diamond A)$$

forallR

$$B \Vdash \Diamond A, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \forall C.((B\mathcal{R}C) \to (C \Vdash \Diamond A))$$

mBoxR

$$B \Vdash \Diamond A, \forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash B \Vdash \Box(\Diamond A)$$

implyR

$$\forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash (B \Vdash \Diamond A) \to (B \Vdash \Box(\Diamond A))$$

mImplyR

$$\forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash B \Vdash (\Diamond A) \to (\Box(\Diamond A))$$

forallR

$$\forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v)))) \vdash \forall t.(t \Vdash (\Diamond A) \to (\Box(\Diamond A)))$$

implyR

$$\vdash (\forall w.(\forall u.(\forall v.(((w\mathcal{R}u) \wedge (w\mathcal{R}v)) \to (u\mathcal{R}v))))) \to (\forall t.(t \Vdash (\Diamond A) \to (\Box(\Diamond A))))$$